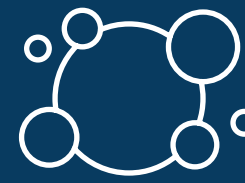


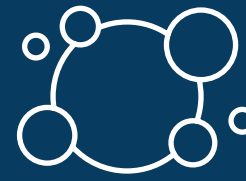
Trace Minerals



TRACE MINERALS AND STANDARD RECOMMENDATIONS										
		RDA	AI	RDA	RDA	RDA	AI	RDA	RDA	AI
Sex	Age	Iron (Fe) (mg·d ⁻¹)	Fluorine (F) (mg·d ⁻¹)	Zinc (Zn) (mg·d ⁻¹)	Copper (Cu) (µg·d ⁻¹)	Selenium (Se) (µg·d ⁻¹)	Manganese (Mn) (mg·d ⁻¹)	Iodine (I) (µg·d ⁻¹)	Molybdenum (Mb) (µg·d ⁻¹)	Chromium (Cr) (µg·d ⁻¹)
Males	14-18	11	3	11	890	55	2.2	150	43	35
	19-30	8	4	11	900	55	2.3	150	45	35
	31-50	8	4	11	900	55	2.3	150	45	35
	51-70	8	4	11	900	55	2.3	150	45	30
	70+	8	4	11	900	55	2.3	150	45	30
Females	14-18	15	3	9	890	55	1.6	150	43	24
	19-30	18	3	8	900	55	1.8	150	45	25
	31-50	18	3	8	900	55	1.8	150	45	25
	51-70	8	3	8	900	55	1.8	150	45	20
	70+	8	3	8	900	55	1.8	150	45	20

Recommended Dietary Allowance (RDA); Adequate Intake (AI); milligrams per day (mg·d⁻¹); micrograms per day (µg·d⁻¹). Adapted from Gropper & Smith (2022) for age categories in physique competitions.

Trace Minerals

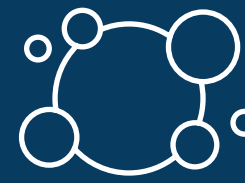


TOLERABLE UPPER INTAKE LEVEL (UL) FOR TRACE MINERALS

Sex	Age	Iron (mg·d ⁻¹)	Fluorine (mg·d ⁻¹)	Zinc (mg·d ⁻¹)	Copper (µg·d ⁻¹)	Selenium (µg·d ⁻¹)	Manganese (mg·d ⁻¹)	Iodine (µg·d ⁻¹)	Molybdenum (µg·d ⁻¹)	Boron (mg·d ⁻¹)	Nickel (mg·d ⁻¹)	Vanadium (mg·d ⁻¹)
Males / Females	14-18	45	10	34	8,000	400	9	900	1,700	17	1.0	–
	19-50	45	10	40	10,000	400	11	1,100	2,000	20	1.0	1.8
	51-70	45	10	40	10,000	400	11	1,100	2,000	20	1.0	1.8
	70+	45	10	40	10,000	400	11	1,100	2,000	20	1.0	1.8

The Tolerable Upper Intake Level (UL) for Chromium has not been established (Russell et al., 2001). The dash (-) denotes that mineral UL was not expressed due to a lack of data, not because Vanadium is safe to consume at any level for this age group. Milligrams per day (mg·d⁻¹); micrograms per day (µg·d⁻¹). Adapted from Gropper & Smith (2022) for age categories in physique competitions.

Trace Minerals

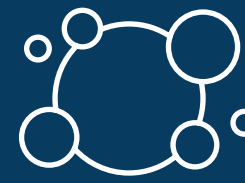


TRACE MINERAL ABSORPTION, FUNCTIONS, AND SYMPTOMS RELATED TO DEFICIENCY OR TOXICITY

	Iron (Fe)	Fluorine (F)	Zinc (Zn)	Copper (Cu)	Selenium (Se)	Manganese (Mn)	Iodine (I)	Molybdenum (Mb)	Chromium (Cr)
Food Sources	Liver, red meat, eggs, clams, oysters, nuts, legumes, green leafy vegetables, dried fruits, broccoli, enriched/whole grains	Milk, egg yolk, seafood, drinking water	Oysters, shellfish, beef, poultry, dairy products (milk, cheese, yogurt), liver, poultry, whole grains, legumes, vegetables, asparagus, spinach	Liver, shellfish, meat, fish, poultry, eggs, bran cereals, nuts, seeds, legumes, broccoli, bananas, avocados	Liver, oysters, tuna, meat, poultry, fish, seafood, cereals, nuts from selenium-rich soil	Wheat bran, legumes, hazelnuts, blueberries, pineapple, shellfish, poultry, meat, nuts, green leafy vegetables, bananas	Iodized salt, saltwater seafood, seaweed, dairy products, liver, eggs, legumes, vegetables	Liver, legumes, meat, poultry, fish, whole-grains, peas, beans	Liver, meat, oysters, cheese, whole-grain products, asparagus, mushrooms, nuts, stainless steel cookware
Co-factor in Enzymatic Roles	Monooxygenases and dioxygenases for amino acid metabolism, carnitine synthesis, and procollagen synthesis; Peroxidases for antioxidant purposes and thyroid hormone synthesis	—	Carbonic anhydrase for acid-base balance and Co2 respiration; Polymerases, kinases, nucleases, transferases; Phosphorylases for nucleic acid synthesis, cell replication, and growth	Superoxide dismutase: antioxidant; Cytochrome-c oxidases for aerobic ATP synthesis; Monooxygenase for norepinephrine synthesis	Glutathione peroxidase as an antioxidant; Iodothyronine 5'-deiodinases for thyroid hormone synthesis	Xylosyl transferase and glycosyl transferase for bone, cartilage, and connective tissue synthesis; Arginase for urea synthesis; Pyruvate carboxylase for carbohydrate metabolism	—	Xanthine oxidoreductase for purine degradation	—

Recommended Dietary Allowance (RDA); Adequate Intake (AI); milligrams per day (mg·d⁻¹); micrograms per day (µg·d⁻¹). Adapted from Gropper & Smith (2022) for age categories in physique competitions.

Trace Minerals



TRACE MINERAL ABSORPTION, FUNCTIONS, AND SYMPTOMS RELATED TO DEFICIENCY OR TOXICITY

	Iron (Fe)	Fluorine (F)	Zinc (Zn)	Copper (Cu)	Selenium (Se)	Manganese (Mn)	Iodine (I)	Molybdenum (Mb)	Chromium (Cr)
Absorption in GI Tract (%)*	iron); 2-10 (non-heme iron)	Not established	20-50	20-50	Not established	Not established	Not established	Not established	<1
Cellular or Physiological Role	O ₂ transport via hemoglobin and myoglobin; forms cytochromes and metalloenzymes; amino acid metabolism; antioxidant; carnitine, collagen, and thyroid hormone synthesis	Promotes bone and teeth formation	Nutrient metabolism, metalloenzymes, promotes protein synthesis, collagen formation, alcohol detoxification, carbon dioxide elimination, tissue repair, energy metabolism, and antioxidant actions	Facilitates iron absorption, oxidative metabolism, synthesis of collagen for connective tissue formation, hemoglobin synthesis, forms cofactor with superoxide dismutase, pigment, neurotransmitters	Antioxidant against hydrogen peroxide and free radicals, forms glutathione peroxidase, thyroid hormone production	Cofactor for energy metabolism enzymes, CNS function, collagen, bone, growth/formation, urea synthesis, and glucose and lipid metabolism	Thyroid hormone (T ₃ and T ₄) synthesis	Cofactor with Riboflavin (B ₂) in CHO and fat metabolism enzymes	It may play a role in augmenting insulin signaling
Deficiency	Anemia, fatigue, impaired work performance; decreased resistance to infection	Dental caries	Poor wound healing, impaired growth, anorexia, abnormal taste/smell, increased risk of infections	Anemia, immune deficiency, bone demineralization, blood vessel abnormalities, impaired immune function	Cardiomyopathy, cancer, heart disease, immune deficiency, abnormal sulfur metabolism, RBC impairment	Poor growth	Enlarged thyroid gland (goiter), impaired RMR	Not established	Glucose intolerance, glucose and lipid metabolism abnormalities



TRACE MINERAL ABSORPTION, FUNCTIONS, AND SYMPTOMS RELATED TO DEFICIENCY OR TOXICITY									
	Iron (Fe)	Fluorine (F)	Zinc (Zn)	Copper (Cu)	Selenium (Se)	Manganese (Mn)	Iodine (I)	Molybdenum (Mb)	Chromium (Cr)
Excess or Toxicity	Hemochromatosis, liver cirrhosis, heart disease, increased risk to infections	Discolored teeth, high doses minimize glycolytic rate	Impaired absorption of both Fe and Cu, increased HDL to LDL cholesterol ratio, anemia, nausea, vomiting, immune deficiency	Nausea, vomiting	Nausea, vomiting, fatigue, hair loss	Weakness, confusion	Deficient thyroid gland activity	Toxic effects are rare	Toxic effects are rare

**The percentage of macromineral that is absorbed through the gut lining. More notable enzymatic roles were selected for each mineral where appropriate. The unabsorbed portion will be excreted with feces. Iron (Fe), Copper (Cu), High-density lipoprotein (HDL), Low-density lipoprotein (LDL), red blood cell (RBC), central nervous system (CNS), resting metabolic rate (RMR). Adapted from Gropper & Smith (2022) and Jeukendrup & Gleeson (2018).*